

Technology-enhanced studio design and creative musicianship: Exploring self-efficacy and motivational moderators

Shiyi Xiao ^{a,b}, Atif Saleem ^{c,*}

^a School of Music, Minzu University of China, Beijing, 100081, China

^b College of Music, Mahidol University, Salaya Campus, Phutthamonthon, Nakhon Pathom, 73170, Thailand

^c School of Education, University of Limerick, Limerick, Ireland

ARTICLE INFO

Keywords:

Technology-enhanced studio design
Creative musicianship
Performance self-efficacy
Teacher autonomy support
Mastery goal orientation

ABSTRACT

This paper examined the possible predictive correlation between technology-enhanced studio design and creative musicianship and the mediating effect of performance self-efficacy in the conditional impact of teacher autonomy support and mastery goal orientation. Validated scales of 418 undergraduate music students in China were collected based on the context of learning in studio and analyzed using PLS-SEM in SmartPLS. Findings showed that technology positively benefited the studio design in relation to creative musicianship and performance self-efficacy. It was also found that, performance self-efficacy was positively related with creative musicianship and this lends credence to the high indirect impact of technology-enhanced studio design on creative musicianship via performance self-efficacy. The moderated mediation analysis showed that a strengthening effect existed between teacher autonomy support as a mediator and mastery goal orientation did not significantly moderate the mediated indirect effect. The performance self-efficacy and creative musicianship had an important variance in the model. The results highlight the importance of technology-enhanced learning settings to support the creative outcomes in music education particularly when teacher autonomy-supportive teaching allows the students to replicate studio affordances into competence-related beliefs and sustained creative interest.

1. Introduction

In education systems in most parts of the world, creativity is no longer a nice-to-have objective in education, that is, an enrichment objective, but an essential capability that comes with innovation, creativity in culture and employability in a creative economy. This shift has been particularly prominent in music education, as there has long been a tradition of teaching music through the exact reproduction of repertoire, sometimes to the cost of the skills and abilities that enable the development of creative musicianship as a whole musical ability (Miller, 2022). Simultaneously, the faster digitization of music learning is transforming the landscape: recording and playback tools, notation and composition programs, digital audio workstations, mobile applications, and interactive platforms have become ubiquitous in most learning environments, altering the music learning process in terms of how learners engage with sound and provide feedback and refine musical concepts. Combining international evidence with recent reviews and empirical studies, one can say that the use of technology-supported music learning can enhance engagement and support creative

outcomes in the case of meaningful use as opposed to meaningless addition (Lam, 2024).

These developments make the studio (broadly defined as the physical and digital learning environment for practice, instruction, and production) far more than the neutral backdrop for action and interaction; it becomes an active pedagogical force. Technology-enhanced studio design can introduce “creative affordances” into the process of everyday learning - ways to easily capture performances, review the detail of those performances over and over, experiment with timbre and structure, compare versions of their work and take feedback and turn it into concrete revision. Such iterative cycles are central to creative development in music as they make it possible to explore, take risks and refine more than one-shot attempts to perform. Evidence from music technology scholarship further suggests that take-up of digital tools by learners is shaped by perceived usefulness, ease and the availability of supportive resources and infrastructure - features that are directly affected by the design and equipment of studios (Waddell & Williamon, 2019). Yet, despite increased interest in the integration of technology, the field remains in need of sharper explanatory models linking the

* Corresponding author.

E-mail address: Atif.Saleem@ul.ie (A. Saleem).

<https://doi.org/10.1016/j.actpsy.2026.107215>

Received 16 January 2026; Received in revised form 24 May 2026; Accepted 4 June 2026

Available online 8 June 2026

0001-6918/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

design of technology-enhanced studios and creative musicianship as an outcome, and not as a disconnected classroom tactic. Addressing this gap is timely for music education research and practice, as institutions worldwide invest in “smart” learning spaces without clear evidence about which design features meaningfully translate into students’ creative musical growth (Lam, 2024).

Mechanism and boundary-condition reasoning is essential for advancing theory in music education because the same learning environment can produce very different outcomes across learners and instructional climates. In explanatory terms, mediators clarify *how* a predictor exerts its influence, while moderators specify *when* and *for whom* that influence is stronger or weaker—together forming a “conditional process” account that is more psychologically realistic than a single direct-effect claim (Namazi & Namazi, 2016). In the proposed relationship between technology-enhanced studio design and creative musicianship, an especially plausible psychological mechanism is performance self-efficacy. Self-efficacy theory argues that perceived capability is a proximal driver of effort, persistence, and strategic action under challenge; when learners believe they can execute demanding tasks, they are more willing to attempt difficult goals, recover after errors, and sustain practice behaviors that support growth (De Brabander & Martens, 2014). From this perspective, a technology-enhanced studio can function as a competence-supporting context by enabling repeated feedback cycles (e.g., recording/review), clearer progress signals, and greater control over practice conditions/resources that can strengthen performance self-efficacy and, in turn, make creative musicianship more attainable because creativity in music often requires experimentation, revision, and tolerating uncertainty.

At the same time, the impact of performance self-efficacy is not likely to be homogenous, and this is why dual moderation is theoretically justified. Teacher autonomy support is one aspect of social context highlighted in the self-determination theory: autonomy-supportive instruction (e.g., choice, recognition of perspective, invitation to self-initiation) tends to foster more adaptive involvement and higher levels of perceived competence while controlling climates tend to inhibit initiative and risk-taking, both features of creative work (Ryan & Deci, 2020). Therefore, teacher autonomy support can conceivably enhance the degree to which technology-enhanced studio design is translated into performance self-efficacy by influencing the construction of students’ interpretations of feedback, tool usage, and effort regulation. Mastery goal orientation based on achievement goal theory further captures stable motivational preferences towards learning, improvement and skill development, mastery-oriented students typically interpret difficulty as informative rather than threatening, they are more likely to turn confidence to exploratory practice and creative production (Midgley et al., 2000). Together, performance self-efficacy, teacher autonomy support, and mastery goal orientation offer a coherent human-centered explanation - connecting attitudes, behavioral self-regulation, and motivational climate - of whether technology-enhanced studios are important, or of the psychological pathway and circumstances under which they are most likely to be most productive in fostering creative musicianship.

China offers a particularly consequential context for studying the effect of learning environments on creative outcomes in music education given that it is at the intersection of the rapid digitalization of education and a highly performance- and evaluation-driven culture of training. National and regional efforts have accelerated the construction of the infrastructure and “smart education” development, the expanding of digital platforms, connectivity, and technology-enabled teaching in the education system, including that of higher education and specialized training contexts (China Education Daily, 2024). At the same time, it is quite common for conservatory and university music students to face a series of high-stake juries, competitive selection and high normative expectations for technical precision, which may increase pressure and limit experimentation (recent evidence indicates the salience of performance anxiety and well-being challenges in music training

environments in China (Bao & Yu, 2025). This mix creates a utilitarian puzzle with a global workforce: the investment in digital and intelligent learning spaces is not going to be necessarily translated into more creative musicianship, when the studios are structured to be mostly efficient and to be evaluated instead of being practiced in an exploratory manner. Other relevant literature on creativity in Chinese music education has also found conflict between past priorities and modern demands to place focus on foreground creativity in teaching and testing (Ge, 2024). It is in this light that it is of specific interest to pay attention to technology enhanced studio design as a predictive of creative musicianship to get actionable evidence to institutions in the context of considering what to emphasize when modernizing studios to leave the question of whether technology exists or not, to whether the studio environment is indeed enabling the possibility of iterative feedback and experimentation and creative risk taking. This research can be applied to policy and practice in other systems in the world that are also investing in the technology-enhanced learning environment and attempting to develop creativity as an output of the educational experience by illuminating this connection in China (Sciences, 2023).

Although there is an increasing interest in creativity and digital transformation in music education, there are a number of gaps that restrict what we can be certain about why learning spaces that are rich in technology promote (or do not promote) creative musicianship. First, the technology-and-creativity literature in music is relatively small and tool-centric; scoping and review research indicates that studies tend to focus on particular technologies or small interventions instead of theorizing the learning environment as a cohesive and designed system that is always conducive to creative development across contexts (Lam, 2024). Second, despite discussing the presence of the concept of technology integration, a gap is reported in the lack of coherent instructional design methodologies and theoretically based models that relate the technology-enabled learning conditions to creativity-relevant musical outcomes, as the studio investment decisions do not correlate with evidence on creative returns (Macrides & Angeli, 2020). Third, music education research has generally considered the studio as a blank slate or operationalized technology as an individual use or attitude, which does not specify the role of technology-enhanced studio design as a structural predictor that influences feedback loops, iteration, and experimentation, processes that form the core of creativity in performance, improvisation, and arranging (Thaçi, 2025). Fourth, the creativity agenda is often discussed in normative terms (i.e., creativity is desirable), yet empirical work still shows persistent tensions between assessment-driven training cultures and creativity-oriented pedagogy, suggesting a need for explanatory models that can reconcile high-performance expectations with creative musicianship outcomes (Yang & Welch, 2023). Fifth, although autonomy-supportive teaching has gained traction in general education and is increasingly studied in music contexts, creativity-focused work still calls for stronger integration of motivational climate into environment-to-outcome models rather than examining instructional variables in isolation from learning-space affordances (Hendriks et al., 2023). Sixth, achievement goal research in music indicates that mastery goal orientation is theoretically relevant and empirically linked to adaptive learning and, in some studies, creativity-related outcomes, yet it is rarely positioned as a boundary condition in models of technology-enabled learning environments and creative musicianship (Tan & Sin, 2020). By addressing these gaps together, the present study advances a more decision-relevant and psychologically grounded account of how technology-enhanced studio design relates to creative musicianship.

2. Literature review

2.1. Self-determination theory as the underpinning lens

Self-Determination Theory (SDT) is a popular macro-theory of human motivation that is used to explain how social contexts determine

the quality of individual motivation and, therefore, their learning, performance and well-being (Ryan & Deci, 2000). Developed out of the background of intrinsic motivation and internalization, SDT was formalized in the mid-1980s and has since grown to be a complete framework applicable in the field of education. One of the key ideas of SDT is that the optimal functioning is determined by the fulfillment of three primary psychological needs, namely, autonomy (the sense of volitional and self-endorsing), competence (the sense of effectiveness), and relatedness (the sense of being connected). The more these needs are supported, the more the learners tend to exhibit intrinsic motivation, extended engagement, adaptive self-regulation, and growth-oriented behaviors; when inhibited, motivation is controlled, less robust, and less conducive to exploration and creativity (Ryan & Deci, 2020). The current study has a good theoretical basis based on SDT since the inherent relationship between technology-enhanced studio design and creative musicianship is motivational and situation specific. The designer studio may be theorized as a need-supportive learning environment: carefully designed tools and infrastructure can enhance perceived competence through facilitating frequent feedback, iterative refinement (e.g., recording review cycles), and indicators of progress, as well as support autonomy by offering learners greater opportunities to engage in self-directed experimentation and creative problem solving. The autonomy support of teachers is a fundamental SDT social-context measure, with a measurement tradition (e.g., Learning Climate Questionnaire) that assesses the degree to which teachers promote choice, respect student views, and endorse self-initiation (Ryan et al., 2025). In this context, SDT provides a logical account of why enabling conditions are likely to foster creative musicianship, a process that cannot be achieved through technical proficiency alone but also requires volitional interest, exploratory practice, and a readiness to take expressive risks. Thus, technology-enhanced studio design may support creative musicianship when it strengthens students' autonomy, competence, and willingness to engage in experimentation during musical learning and performance.

2.2. Hypotheses

Learning environments are increasingly considered to be active drivers of creative outcomes as opposed to a kind of backdrop. Across fields of education, evidence suggests that well-designed learning spaces, particularly those that allow for interaction, iteration and learner agency support more in-depth engagement and higher order outcomes including the development of creativity (Davies et al., 2013). From the point of view of education management, this changes the focus from teaching tactics to institutional choices: the way in which resources, infrastructure and space configurations are planned, can have material effects on the learning experiences of students and the types of behaviors made easy and difficult in a given setting. Contemporary research on learning-space design focuses on the ways in which technology-rich learning spaces can support more active, collaborative and experiential learning but benefits are dependent on how the design can match the intended pedagogy and ensure the conditions of support are usable and reliable respectively (Papaioannou et al., 2023). In other words, though, "technology present" is not equivalent to "technology enabling" - the educational value comes when the design of the space reduces friction for meaningful practice routines (feedback, revision, experimentation) and supports sustained engagement.

This logic is particularly in focus in music education where creative musicianship is formed through cycles of exploration, feedback, and refinement in the areas of improvisation, composition/arranging, and expressive performance. Recent syntheses of technology-enhanced creativity in music education demonstrate that digital tools can be supportive of creative outcomes when they are used in ways that promote experimentation and iterative making instead of digitizing traditional routines that are about reproduction (Lam, 2024). Studio-based work also provides a further example of the importance of feedback and

revision processes as digital technology allows for more complex forms of performance capture, replay, and targeted revision by studio learning (Hamond et al., 2020). Accordingly, technology-enhanced studio design, through access to recording/review workflows, composition/production tools, and reliable support, should increase opportunities for students to generate, test, and refine musical ideas, thereby strengthening creative musicianship. Interpreted through Self-Determination Theory, such studios can function as need-supportive contexts that facilitate autonomous engagement and perceived competence, conditions that are conducive to creative growth. Therefore, we propose:

H1. Technology-enhanced studio design is positively related to creative musicianship.

The explanations that better explain environmental elements are those that determine the psychological belief in which a setting is made actionable to the learners (Yim, 2023). Self-efficacy grows in general education and training research when contexts involve recurring possibilities of mastery experiences, when information about performance is available at the right time, and when circumstances render improvement noticeable and manageable (Waddington, 2023). Technology-enhanced studio design can make these efficacy-building conditions institutional, embedded in music education through recording/review routines, easy access to digital tools, and reliable support that minimizes the barriers of deliberate practice and sequential refinement. The studies on the application of technology in music learning indicate that the perceived usability and perceived learning value significantly influence the level to which musicians use digital tools, implying that properly designed studio infrastructure can be translated into an increased number of and quality technology-mediated practice experiences (Beattie et al., 2016). In this regard, the design of the studio equipped with technology should have a positive relationship with performance self-efficacy (Zelenak, 2015).

Generally speaking, it can be stated that self-efficacy is always connected to persistence, strategic self-controlling, and performance in a challenge setup since it determines the interpretation of difficulty and the persistence when one is hit-back (Li, 2023). Creativity studies also show that efficacy beliefs have a significant relationship with creative performance; a meta-analysis shows that creative self-efficacy and creative performance are medium-sized correlated in the studies (Haase et al., 2018). Learning creative musicianship in music involves experimentation, revision, and expressive risk-taking, which is more likely to happen as students feel they can perform to the demands, and will recover should they make a mistake. In line with the previous studies on music performance self-efficacy, the stronger performance self-efficacy must correlate with greater creative musicianship (Zelenak, 2015).

The theoretical suitability of mediation is that the design of studios is likely to influence creative musicianship by enhancing beliefs related to competence that maintain the exploratory practice (Liu, 2025). Technology-supported studio design has the potential to create performance self-efficacy, through facilitating repeated, competence-relevant feedback loops (e.g. capture -> review -> adjust -> reattempt), making progress more visible and manageable, which lies at the core of efficacy formation (Zimu, 2024). Such environments may also support competence need satisfaction, which facilitates better-quality engagement and development-focused results that are compatible with creative musicianship, according to an SDT lens.

H2. There is a positive relationship between technology-enhanced studio design and performance self-efficacy.

H3. Performance self-efficacy and creative musicianship have a positive relationship.

H4. Performance self-efficacy is a mediating variable between technology enhanced studio design and creative musicianship.

The use of conditional process theorizing is becoming more and more encouraged when educational interventions and learning conditions do

not equally work in instructional climates and orientations of learners (Hayes & Rockwood, 2020). Moderated mediation tests the moderation of an indirect effect (technology-enhanced studio design to performance self-efficacy and to creative musicianship) by a moderator, which provides a more realistic description of what works, who, and under which conditions (Hayes, 2017). In a broad perspective, SDT claims that contextual supports influence motivation and competence-related functioning because they satisfy (or frustrate) basic psychological needs, in particular, autonomy and competence (Tang et al., 2020). It means that, with the same technology tools, learners, whose social climate is autonomy-supportive, can produce different psychological outcomes. Teacher autonomy support, in the case of music education, cannot be simply a nice person attribute, as students can perceive feedback as informational or controlling, they may feel safe to experiment, and they may stay in the process of iterative revision that is oftentimes antecedent to creative production. The recent studies on music-education have clearly indicated that autonomy-supportive teaching is explicitly associated with processes and outcomes related to creativity, which confirms the idea that autonomy support may be a contextual amplifier to creative development (Hendriks et al., 2023). The teacher autonomy support, thus, should enhance the indirect impact of technology-enhanced studio design on creative musicianship by boosting the degree to which the studio affordances are converted to performance self-efficacy (i.e., higher first-stage influence with greater autonomy support) (Ryan & Deci, 2020).

The second boundary condition is related to the motivation orientation of learners to growth. In terms of general achievement motivation, mastery goal orientation indicates the desire to enhance competence by working hard and perceiving challenge as diagnostic and improvement oriented as opposed to dangerous. There is evidence that mastery goal orientation has positive relationships with self-regulated learning strategies and adaptive engagement, particularly in learning conditions that involve persistence and refinement through trial and error (Beik & Cho, 2024). The music-studio-setting context puts creative

musicianship on repetition, trying out alternatives, tolerating imperfect drafts, and revising interpretations, and one would expect the translation of performance self-efficacy into creative musicianship to be more pronounced in mastery-oriented students who tend to put confidence in exploration as opposed to error avoidance. In the music-studio context, creative musicianship hinges on repeated experimentation, trying alternatives, tolerating imperfect drafts, and revising interpretations, so the translation of performance self-efficacy into creative musicianship should be stronger among mastery-oriented students who are more willing to invest confidence into exploration rather than only error avoidance. This aligns with SDT's emphasis on high-quality, growth-oriented engagement: when competence beliefs are high, mastery-oriented students are especially likely to enact those beliefs through sustained, exploratory practice that supports creativity. Therefore, mastery goal orientation is expected to condition the mediated pathway by strengthening the latter part of the mechanism (i.e., a stronger performance self-efficacy → creative musicianship link), producing a stronger conditional indirect effect at higher mastery goal orientation (Edwards & Konold, 2020).

H5. The indirect effect of technology-enhanced studio design on creative musicianship via performance self-efficacy is stronger when teacher autonomy support is higher.

H6. The indirect effect of technology-enhanced studio design on creative musicianship via performance self-efficacy is stronger when mastery goal orientation is higher.

A conceptual model is presented in Fig. 1 to summarize the hypothesized direct, indirect, and conditional (moderated mediation) pathways. The model is grounded in the premise that learning environments can function as motivationally meaningful contexts that support competence-relevant beliefs and agentic engagement, thereby facilitating creative outcomes. In self-determination theory, contexts that support autonomy and competence are expected to strengthen internalization, persistence, and exploratory engagement, which are

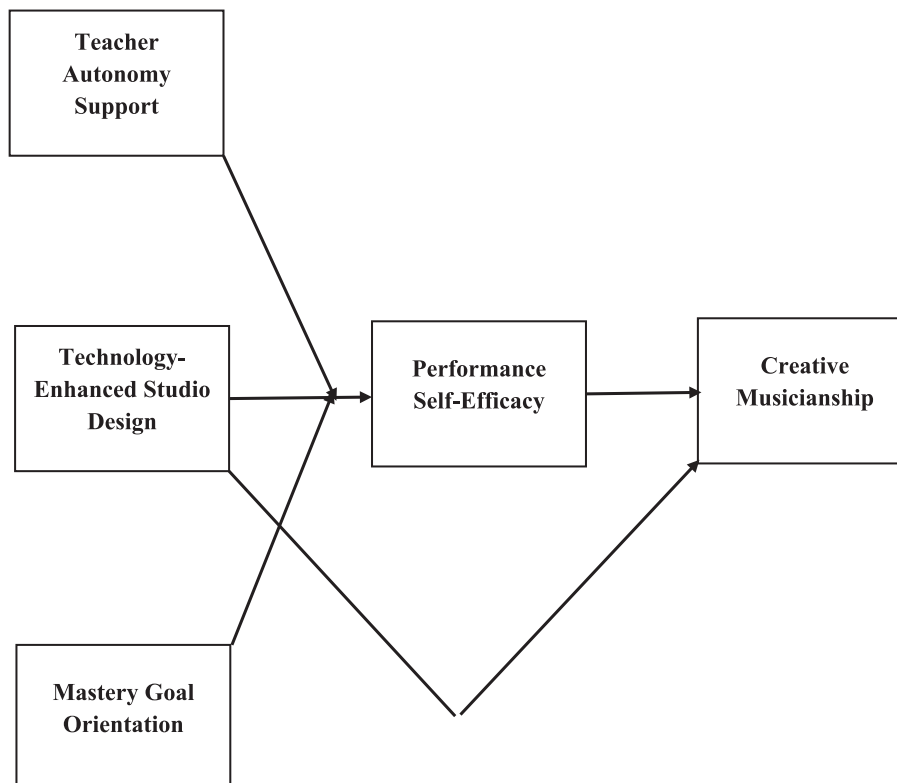


Fig. 1. Conceptual model.

central to creative skill development.

3. Methods

3.1. Research design

The study employed a cross-sectional survey design, allowing for simultaneous assessment of predictors, mediators and moderators among undergraduate music students. Cross-sectional conditional process designs are suitable for testing theoretically specified mediation and moderation patterns when longitudinal data are unavailable (Namazi & Namazi, 2016). Although cross-sectional data limit causal inference, the theoretical model was grounded in established frameworks, self-determination theory and achievement goal theory, and the design captured relationships among technology-enhanced studio environments, psychological mediators, and creative outcomes.

3.2. Participants

Participants were undergraduate and/or higher-education music learners in China ($N = 418$). Table 1 reports participant characteristics (gender, age categories, and year of study). The sample size exceeded typical minimum requirements for structural equation modeling estimation and provided stable estimates for the specified measurement and structural parameters, including interaction terms.

As shown in Table 1, the final sample ($N = 418$) included slightly more female (53.6%) than male (44.0%) respondents, with most participants aged 20–21 years (43.5%). Students were distributed across year levels, with Year 2 (32.1%) and Year 3 (27.8%) forming the largest groups. About half of respondents were enrolled in music performance

Table 1
Socio-demographic characteristics of respondents ($N = 418$).

Characteristic	Category	Frequency (n)	Percent (%)
Gender	Male	184	44.0
	Female	224	53.6
	Prefer not to say	10	2.4
Age (years)	18–19	96	23.0
	20–21	182	43.5
	22–23	104	24.9
	24+	36	8.6
Year of study	Year 1	102	24.4
	Year 2	134	32.1
	Year 3	116	27.8
	Year 4	66	15.8
Major/track	Music performance	212	50.7
	Music education	118	28.2
	Composition/arranging	54	12.9
	Other music-related	34	8.1
Primary specialization	Piano	138	33.0
	Vocal	96	23.0
	Strings	62	14.8
	Wind/Brass	46	11.0
	Percussion	30	7.2
	Traditional Chinese instruments	46	11.0
Formal training length (music study)	<5 years	64	15.3
	5–7 years	126	30.1
	8–10 years	142	34.0
	11+ years	86	20.6
Weekly practice time	<5 h	84	20.1
	5–9 h	142	34.0
	10–14 h	118	28.2
	15+ hours	74	17.7
Prior use of music technology (apps/DAW/recording)	Low	112	26.8
	Moderate	196	46.9
	High	110	26.3

tracks (50.7%), and piano (33.0%) and vocal (23.0%) were the most common specializations. Most students reported moderate prior use of music technology (46.9%) and practiced 5–9 h per week (34.0%).

3.3. Procedure

Data were collected using a self-administered questionnaire that assessed perceived technology-enhanced studio design, performance self-efficacy, teacher autonomy support, mastery goal orientation, and creative musicianship. Participation was voluntary, and respondents were informed about the study purpose, confidentiality, and their right to discontinue at any time prior to starting the survey (Li et al., 2024). To reduce common method bias risk in single-source survey designs, procedural remedies were applied (e.g., emphasizing anonymity, minimizing evaluation apprehension, and embedding attention checks), consistent with recommendations for cross-sectional self-report research (Ahmad, Samad, & Han, 2024). As an additional statistical safeguard, a marker-variable approach was included in the analysis to evaluate whether common method variance was likely to meaningfully inflate observed relationships.

3.4. Ethical considerations

This study received ethical approval from the Academic Ethics Committee of the School of Music, Minzu University of China (Ref. No. MUC-MUS-ETH-2025-0720; approval date: June 06, 2025). The Committee reviewed and approved the study objectives, research design, instruments, sampling procedures, and data management plan. Participation was voluntary, and informed consent was obtained from all participants before data collection. Participants' privacy and confidentiality were protected throughout the study; no names or student IDs were collected in the main survey, and all findings were reported only in aggregate form. The study was classified as involving no more than minimal risk, and all data were stored securely and used solely for academic research and publication purposes (Ahmad, Samad, & Mahmood, 2024; Wang et al., 2024).

3.5. Measures

All constructs were measured using established instruments with careful contextual adaptation for studio-based music learning. Unless otherwise noted, items were rated on a five-point Likert scale (1 = strongly disagree, 5 = strongly agree). To ensure measurement rigor and minimize later-stage review concerns, the instrument development process followed best practices for survey research and scale adaptation. First, scales with strong prior usage and psychometric support were selected for each construct. Second, item wording was adapted minimally to reflect music-studio and lesson contexts (e.g., “class” or “learning environment” phrasing was reframed to “studio,” “lesson,” or “practice”) while preserving the original conceptual meaning. Third, an expert review panel (music education faculty and research-methods specialists) evaluated item relevance, clarity, and representativeness; minor edits were made for readability and context-fit. Fourth, cognitive interviews with a small group of music students were used to verify comprehension and reduce ambiguity. Fifth, a pilot test was conducted to check item functioning (e.g., ceiling/floor effects and item-total patterns) before full administration. Because the questionnaire was administered in Chinese, all scale items were translated from English into Chinese using a forward-translation procedure and then independently back-translated into English. Any discrepancies between the original and back-translated versions were reviewed and reconciled by the research team to ensure conceptual equivalence and contextual clarity. In the main study, procedural remedies were implemented to strengthen response quality (e.g., anonymity assurance, separation of construct blocks, mixed item ordering within blocks, and attention checks), and low-quality responses were screened out using pre-

specified rules (e.g., excessive missingness, failed attention checks, straight-lining, and implausibly fast completion).

Technology-enhanced studio design was measured with eight items capturing students' perceptions of the studio's technology-enabled affordances and support conditions for practice and learning, adapted from Waddell and Williamon (2019), with conceptual anchoring from facilitating conditions in Venkatesh et al. (2003) study. Sample items include "The studio provides the digital tools (e.g., apps/software/recording tools) I need for practice" and "The studio setup makes it easy to record and review my performance for improvement." Teacher autonomy support was assessed with six items adapted from the Learning Climate Questionnaire (Williams et al., 2006), reflecting the extent to which instructors provide choice, acknowledge student perspectives, and support self-initiation. Sample items include "I feel that my teacher provides me with choices and options in how I learn/practice" and "I feel that my teacher listens to how I would like to do things." Mastery goal orientation was measured with four items adapted from the mastery goal subscale of the Patterns of Adaptive Learning Scales (Midgley et al., 2000). Sample items include "One of my goals in this class is to learn as much as I can" and "It is important to me that I improve my skills in this course."

Performance self-efficacy (mediator) was measured using five items adapted from music performance self-efficacy research (Zelenak, 2015), capturing students' confidence in preparing for and executing performances and managing performance demands. Sample items include "I can learn difficult pieces successfully" and "I can perform well in front of an audience." Creative musicianship (dependent variable) was measured with seven items aligned with domain creativity in artistic/performance contexts and adapted to music-specific behaviors (Kaufman, 2012). Items captured key facets of creative musicianship such as improvisation, composing/arranging, and expressive originality. Sample items include "I can improvise musical ideas that feel original to me" and "I can arrange or adapt music in creative ways (e.g., changing style, rhythm, or harmony)." All items were coded so that higher values reflected higher levels of the construct; any negatively worded items included during piloting were reverse-coded prior to analysis.

3.6. Data analysis

The hypothesized model was estimated using partial least squares structural equation modeling (PLS-SEM) in SmartPLS. PLS-SEM was selected for four reasons. First, the primary analytic goal was explanation and prediction of key target constructs (performance self-efficacy and creative musicianship) via maximization of explained variance and identification of practically meaningful drivers, which is consistent with the causal-predictive orientation of PLS-SEM (Chen et al., 2026; Imran et al., 2026). Second, PLS-SEM is frequently recommended when models include multiple mediating and moderating paths and when researchers prioritize robust estimation with bootstrapped inference over global covariance-based fit emphasis (Ahmad et al., 2026; Liu et al., 2026). Third, the model included latent interaction terms and conditional indirect effects; PLS-SEM provides established procedures for estimating moderating effects and moderated mediation in composite-based estimation frameworks (Jiang et al., 2025). Fourth, because cross-sectional field survey data often deviate from strict multivariate normality, PLS-SEM's distributional flexibility supported stable estimation using nonparametric bootstrapping. Measurement model evaluation followed standard recommendations: indicator loadings and collinearity diagnostics (VIF), internal consistency reliability (composite reliability), convergent validity (average variance extracted), and discriminant validity using the HTMT criterion. Structural model evaluation included collinearity checks, bootstrapped path coefficients, effect sizes (f^2), and explained variance (R^2), consistent with established PLS-SEM reporting guidance.

4. Results

4.1. Measurement model assessment

Before hypothesis testing, we evaluated the measurement model for indicator reliability, internal consistency reliability, convergent validity, and collinearity. After the first mention of Creative musicianship (CM), Mastery goal orientation (MGO), Performance self-efficacy (PSE), Teacher autonomy support (TAS), and Technology-enhanced studio design (TESD), we report results using abbreviations. As shown in Table 2, standardized outer loadings were generally strong (most ≥ 0.700). A small number of indicators fell slightly below 0.700 (e.g., CM3 = 0.699; MGO3 = 0.666; MGO4 = 0.647; PSE2 = 0.601; TAS4 = 0.662; TAS5 = 0.660; TAS6 = 0.680; TESD3 = 0.681), but they were retained because composite reliability and AVE remained acceptable and the items were important for content coverage. Convergent validity was supported, with AVE values ranging from 0.537 to 0.575, exceeding the 0.500 benchmark. Internal consistency reliability was also satisfactory, as composite reliability values ranged from 0.831 to 0.905 which were well above the commonly accepted threshold of 0.700 (Ahmad, Ahmad, Siddique, et al., 2024). Finally, indicator collinearity was within acceptable limits: all VIF values were below 5.000 (maximum = 4.660), suggesting that multicollinearity was not a concern (Hair & Alamer, 2022; Kock, 2015).

Fig. 2 presents the reflective indicators for TESD, PSE, CM, TAS, and MGO with their standardized outer loadings. Most loadings are above 0.700, indicating acceptable indicator reliability, and the endogenous constructs show moderate explanatory power ($R^2 = 0.389$ for PSE; $R^2 = 0.454$ for CM).

Discriminant validity was supported using both criteria. The Fornell–Larcker results show that the square root of AVE (diagonal values)

Table 2
Measurement model results (outer loadings, VIF, composite reliability, and AVE).

Construct	Items	Item Loadings	VIF	CR	AVE
Creative musicianship (CM)	CM1	0.813	3.666	0.904	0.575
	CM2	0.751	2.712		
	CM3	0.699	2.458		
	CM4	0.717	2.994		
	CM5	0.746	3.413		
	CM6	0.746	1.798		
	CM7	0.825	3.792		
Mastery goal orientation (MGO)	MGO1	0.814	1.902	0.831	0.555
	MGO2	0.834	1.978		
	MGO3	0.666	1.359		
	MGO4	0.647	1.365		
Performance self-efficacy (PSE)	PSE1	0.776	4.608	0.867	0.567
	PSE2	0.601	1.292		
	PSE3	0.801	3.599		
	PSE4	0.795	3.592		
	PSE5	0.775	4.590		
Teacher autonomy support (TAS)	TAS1	0.837	4.166	0.873	0.537
	TAS2	0.849	4.237		
	TAS3	0.682	2.416		
	TAS4	0.662	2.453		
	TAS5	0.660	1.853		
	TAS6	0.680	1.937		
Technology-enhanced studio design (TESD)	TESD1	0.750	4.286	0.905	0.544
	TESD2	0.722	2.012		
	TESD3	0.681	1.852		
	TESD4	0.800	3.255		
	TESD5	0.721	3.475		
	TESD6	0.718	4.660		
	TESD7	0.707	3.551		
	TESD8	0.794	3.782		

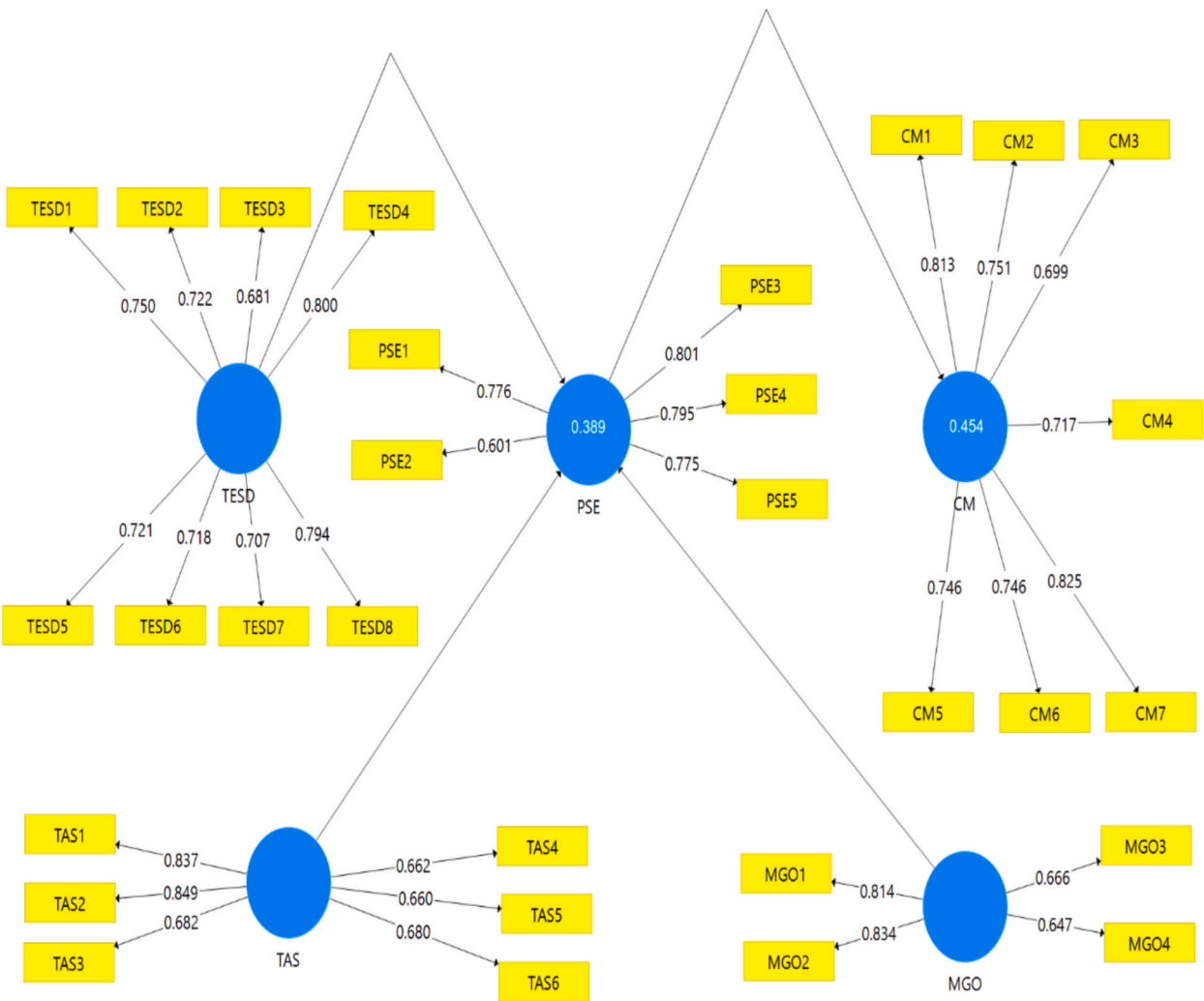


Fig. 2. Measurement model with standardized outer loadings and explained variance (R^2).

exceeded all corresponding inter-construct correlations for CM, MGO, PSE, TAS, and TESD, indicating adequate construct distinctiveness. Consistent with this, all HTMT ratios were below the conservative 0.85 threshold (highest = 0.707), further confirming discriminant validity (Ahmad, Ahmad, Lewandowska, et al., 2024). As shown in Table 3A, the square root of each construct's AVE (on the diagonal) exceeded all inter-construct correlations, supporting discriminant validity. Table 3B presents HTMT values; all were below the recommended threshold of 0.85, indicating that constructs were empirically distinct.

To further assess common method bias, we used a marker variable technique, which is recommended as a statistical remedy alongside procedural controls when self-report data are collected from a single source. Following the correlational marker-variable approach (Lindell & Whitney, 2001), we included an auxiliary construct expected to be

Table 3A
Fornell–Larcker Criterion.

Constructs	CM	MGO	PSE	TAS	TESD
CM	0.758				
MGO	0.498	0.745			
PSE	0.402	0.396	0.753		
TAS	0.436	0.344	0.417	0.733	
TESD	0.304	0.442	0.491	0.239	0.738

Note. Bold diagonal values are the square roots of AVE. Off-diagonal values are inter-construct correlations.

Table 3B
Heterotrait–Monotrait Ratio (HTMT).

Constructs	CM	MGO	PSE	TAS	TESD
CM	–				
MGO	0.610	–			
PSE	0.681	0.616	–		
TAS	0.496	0.669	0.683	–	
TESD	0.707	0.499	0.667	0.417	–

theoretically unrelated to the focal variables: Campus Cafeteria Satisfaction (CCS). CCS taps students' satisfaction with food quality, price, and cleanliness of campus dining services, conceptually distant from this study's variables. As shown in Table 4, CCS exhibited very low correlations with all substantive constructs ($|r| \leq 0.15$), suggesting that common method variance is unlikely to meaningfully inflate the relationships among the focal variables. This pattern aligns with recommendations that marker–substantive correlations should be small if common method variance is not a serious threat.

4.2. Structural model (hypotheses testing)

SmartPLS (PLS-SEM) was used to test the structural model because it supports prediction-oriented modeling and is well suited for models that include mediation and moderated mediation, with significance evaluated via nonparametric bootstrapping and bias-corrected confidence

Table 4
Descriptive statistics and correlations with the marker variable (CCS).

Variable	Mean	SD	1	2	3	4	5	6
1. CM	3.76	0.692	1.00					
2. MGO	3.92	0.660	0.500	1.00				
3. PSE	3.61	0.721	0.402	0.400	1.00			
4. TAS	3.85	0.714	0.442	0.344	0.422	1.00		
5. TESD	3.68	0.742	0.303	0.443	0.490	0.243	1.00	
6. CCS (marker)	3.45	0.800	0.090	0.123	0.083	0.151	0.111	1.00

Note. CM = Creative musicianship; MGO = Mastery goal orientation; PSE = Performance self-efficacy; TAS = Teacher autonomy support; TESD = Technology-enhanced studio design; CCS = Campus Cafeteria Satisfaction (marker variable). All correlations involving CCS are small ($|r| \leq 0.15$), indicating minimal shared variance with the focal constructs.

intervals. To improve clarity, the hypothesis-testing results are presented in two separate tables. Table 6A reports the direct structural paths, while Table 6B presents the indirect and conditional indirect effects. As shown in Table 6A, the direct paths were significant and positive: TESD → CM (H1) was supported ($\beta = 0.428$, $t = 12.340$, $p < .001$), TESD → PSE (H2) was supported ($\beta = 0.438$, $t = 11.867$, $p < .001$), and PSE → CM (H3) was supported ($\beta = 0.410$, $t = 10.482$, $p < .001$). The indirect effect TESD → PSE → CM (H4) was also significant ($\beta = 0.179$, $t = 9.036$, $p < .001$), indicating mediation. For conditional indirect effects, TAS significantly moderated the mediated relationship (H5 supported; $\beta = 0.043$, $t = 2.266$, $p < .05$), whereas MGO did not significantly moderate the mediated relationship (H6 not supported; $\beta = 0.019$, $p = .106$; CI included zero).

Regarding explanatory power (Table 7), the model accounted for 0.454 of the variances in CM and 0.389 in PSE (R^2), indicating moderate explanatory power for behavioral survey research using PLS-SEM. Effect sizes (f^2) as reported in Table 8 were interpreted using established benchmarks (0.02 = small, 0.15 = medium, 0.35 = large). The f^2 values suggest that TESD → PSE (0.192) and PSE → CM (0.197) were medium effects, TAS → PSE (0.143) was small-to-medium, and MGO → PSE (0.071) was small, indicating that TESD and PSE were the most practically meaningful drivers in the model.

Fig. 3 shows positive, significant direct paths from TESD to PSE and CM, and from PSE to CM, supporting the mediation mechanism. The interaction TESD×TAS strengthens the pathway via PSE, whereas TESD×MGO shows a weaker conditional effect, consistent with hypothesis testing.

The simple-slope plots illustrate how TESD relates to PSE at low (−1 SD), mean, and high (+1 SD) levels of each moderator. In the TESD×MGO (Fig. 4), the lines are nearly parallel, indicating that changes in MGO do not meaningfully alter the TESD→PSE slope, consistent with the non-significant moderated mediation result (H6 not supported). In contrast, the TESD×TAS (Fig. 5) shows clearly diverging slopes, indicating that higher TAS strengthens the positive TESD→PSE relationship, aligning with the significant conditional indirect effect (H5 supported).

5. Discussion

This study examined whether technology-enhanced studio design predicts creative musicianship and whether performance self-efficacy mediates this relationship under specific motivational conditions (teacher autonomy support and mastery goal orientation). The results

Table 6A
Direct effects and hypothesis testing.

Hypothesized path	β	SD	t	p	95% CI	Decision
H1: TESD → CM	0.43	0.04	12.34	< 0.001	[0.35, 0.49]	Supported
H2: TESD → PSE	0.44	0.04	11.87	< 0.001	[0.36, 0.51]	Supported
H3: PSE → CM	0.41	0.04	10.48	< 0.001	[0.34, 0.49]	Supported

Table 6B
Indirect and conditional indirect effects.

Hypothesized path	β	SD	t	p	95% CI	Decision
H4: TESD → PSE → CM	0.18	0.02	9.04	< 0.001	[0.14, 0.22]	Supported
H5: TESD × TAS → PSE → CM	0.04	0.02	2.27	0.02	[0.01, 0.06]	Supported
H6: TESD × MGO → PSE → CM	0.02	0.02	1.19	0.11	[−0.04, 0.02]	Not supported

Note. TESD = Technology-enhanced studio design; PSE = Performance self-efficacy; CM = Creative musicianship; TAS = Teacher autonomy support; MGO = Mastery goal orientation.

Table 7
Explained variance (R^2).

Endogenous construct	R^2
Creative musicianship (CM)	0.454
Performance self-efficacy (PSE)	0.389

Note. The model explained 45.4% of the variance in creative musicianship and 38.9% of the variance in performance self-efficacy.

Table 8
Effect sizes (f^2).

Relationship	f^2	Interpretation
PSE → CM	0.197	Medium
TESD → PSE	0.192	Medium
TAS → PSE	0.143	Small–medium
MGO → PSE	0.071	Small

Note. Effect-size interpretation follows conventional benchmarks: 0.020 = small, 0.150 = medium, 0.350 = large.

support a positive association between technology-enhanced studio design and creative musicianship, and they indicate that performance self-efficacy is a meaningful psychological mechanism linking studio design to creative outcomes. These results are consistent with broader evidence that learning environments supportive of autonomy and competence can strengthen adaptive motivation and engagement, which in turn supports developmentally important outcomes (Davies et al., 2013; Ryan & Deci, 2020). One of the significant contributions is the discovery that performance self-efficacy is a partial account of the relationship between technology-enhanced studio design and creative musicianship. Self-efficacy beliefs in music learning are associated with the confidence of learners in preparation, performance implementation and perseverance in the face of challenge, and confirmed literature in music education demonstrates that self-efficacy is quantifiable and significantly connected with performance-related processes (Zelenak, 2015). Beyond that, research synthesis suggests that self-efficacy beliefs, particularly, creative self-efficacy, are medium-sized correlated

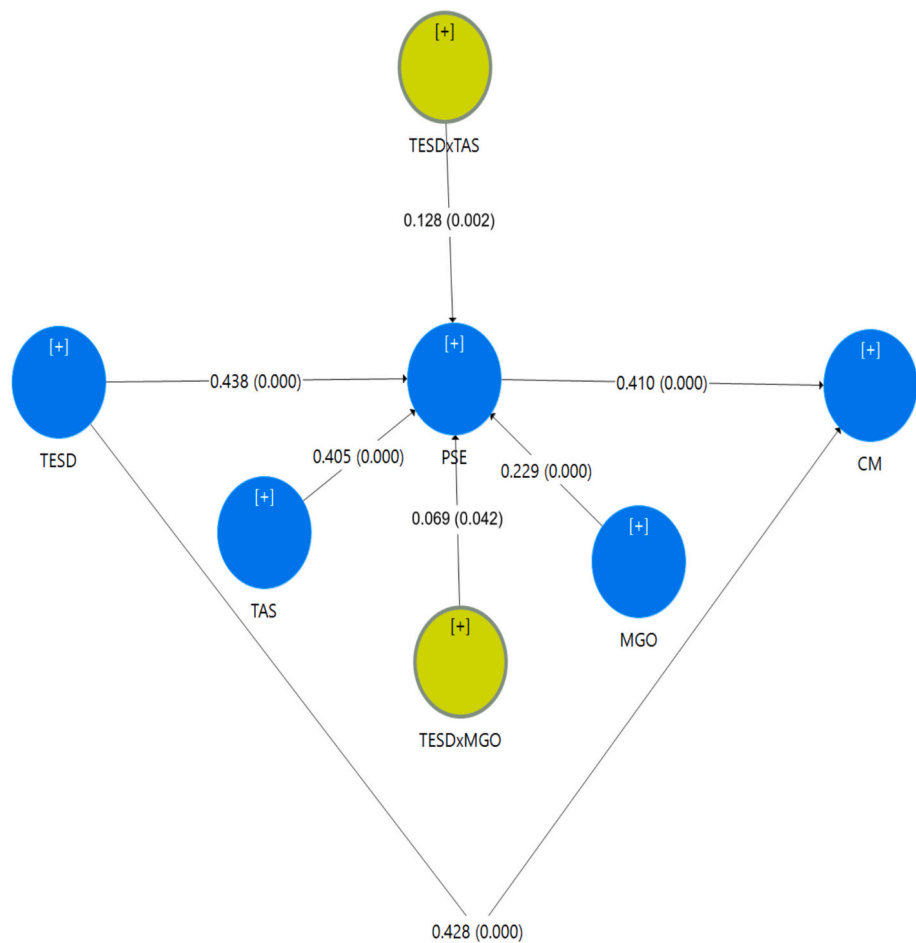


Fig. 3. Structural model with standardized path coefficients (β) and p values, including moderated mediation paths.

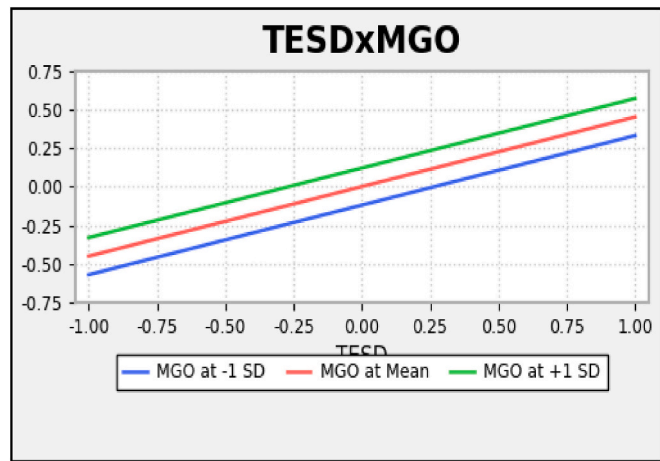


Fig. 4. Simple slope analysis for the TESD $\times$ MGO interaction predicting PSE (non-significant moderation pattern).

with creativity measures, although there is substantial variation in these relationships across creativity measures (Haase et al., 2018). Taken with caution, the current findings can be correlated with the concept that technology-driven studios can assist students in converting practice opportunities into competence beliefs that help them to engage in exploration and creativity.

The conditional process results also indicate that technology-enhanced studio design to creative musicianship through performance

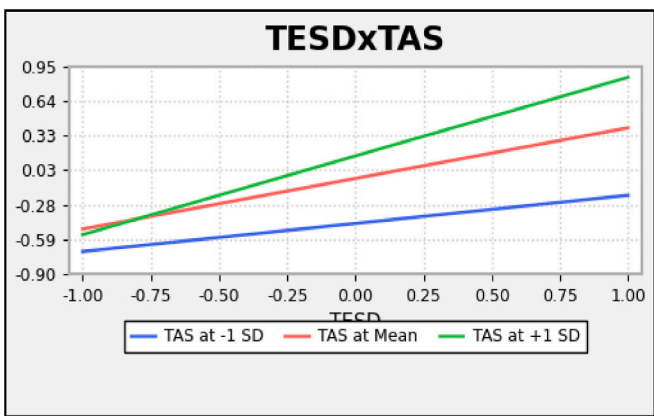


Fig. 5. Simple slope analysis for the TESD $\times$ TAS interaction predicting PSE (significant moderation pattern).

self-efficacy via teacher autonomy support is an indirect connection. Theoretical consistency of this pattern is that the autonomy-enhancing teaching practices (e.g., providing meaningful choices, considering the viewpoint of learners, and encouraging self-initiation) are defining of need-supportive settings as outlined in the self-determination theory and operationalized in the context of music education studies as a means of facilitating learner agency, engagement, and creative growth (Xu & Li, 2025). Notably, the current evidence is cross-sectional meaning that the results are only compatible with (but not conclusive evidence of) the existence of a causal mechanism whereby autonomy-supportive

instruction enhances the advantages of technology-rich practice settings. In contrast, mastery goal orientation did not significantly moderate the mediated pathway in this sample. One interpretation is that technology-enhanced studio design may provide sufficiently structured feedback and practice affordances to support performance self-efficacy across a range of learner goal orientations, reducing the incremental value of mastery-oriented differences in this context. Another interpretation is measurement- and context-dependent. Meta-analytic evidence shows that mastery and mastery-approach goals are generally positively associated with academic self-efficacy, but the strength of this association varies across countries, measurement instruments, learning domains, and the degree of match between goal-orientation and self-efficacy measures (Huang, 2016). Similarly, meta-analytic evidence from online learning contexts suggests that goal-orientation effects may differ across educational and cultural settings (Beik & Cho, 2024). Therefore, future research should test mastery goal orientation as a boundary condition using longitudinal or experimental designs and consider additional moderators, such as performance anxiety, prior technology experience, or teacher feedback style.

Regarding SDT, the results reveal that technology-enhanced studio design is an operation of a need-relevant condition that is primarily competence facilitating by allowing conditions of high levels of feedback and visibility of improvement. In case of competence support, students report higher self-efficacy score of performance that has been linked to more creative musicianship owing to the fact that students are more willing to persist, experiment and refine musical thoughts. Teacher autonomy support also enhances this mechanism in promoting a climate of volition and psychological safety that can assist students in perceiving feedback as informational instead of controlling, to maintain higher-quality engagement. The non-significant moderation by mastery goal orientation implies that contextual need support can dominate some dispositional orientations to explain the translation of studio affordances into creative product.

5.1. Theoretical implications

The article fills theory gaps in music education by combining (i) design of learning environment, (ii) motivational theory focusing on psychological needs, and (iii) mechanism-plus-boundary-condition modeling into one empirical system. Studies of creative learning settings highlight that creative-supportive contexts are recurring in terms of flexibility of resources, timely feedback, learner autonomy, and respectful teacher-learner relationships (Davies et al., 2013). In the context of music education, syntheses and empirical studies show that technology can facilitate creativity when it allows experimentation, iteration, and production tools that are available to it, and is not a mere efficiency tool (Lam, 2024; Waddell & Williamson, 2019). The current results introduce a psychologically defined pathway, performance self-efficacy, by which such design-related affordances can have their effects. The moderated mediation findings point out that the effects of technology are not necessarily technological; they seem to be dependent on motivational climate, particularly, autonomy support. This is consistent with research that teaching autonomy-supportive can be instructed and can facilitate better student results in domains (Xu & Li, 2025) and with music-specific studies highlighting that teacher interactions and behaviors can be used to support creative autonomy in music classrooms (Hendriks et al., 2023; Wang, Gao, Sun, et al., 2025). The cross-domain studies of AI-assisted learning also indicate that the role of teacher support is still relevant in technology-intensive learning, which, again, supports the overall argument that human support systems mediate the influence of technology on education (Wang, Gao, Wang, et al., 2025; Wang & Guo, 2026).

Our results generalize self-determination theory and achievement goal theory to the technology enhanced music education environment. The autonomy supportive climates foster autonomous motivation and perceptions of competence that enhance the effect of environmental

affordances on self-efficacy. Mastery goal orientation, which linked challenges with opportunities to develop, also increased the conversion of technology enhanced studio design to performance self-efficacy. These findings have instilled conditional process models that the environmental design interacts with the social and motivational factors. In addition, the positive association between PSE and creative musicianship supports the hypothesis that self-efficacy is a proximal motivational force of creative performance. The study puts technology enhanced studio design in a wider psychological context, therefore offering a more refined perspective on the interaction of technology and pedagogy in influencing creative results.

5.2. Practical implications

The results provide practical recommendations to music departments aiming to develop creative musicianship based on environment and pedagogy. Technology-enhanced studio design should be a strategic educational investment by institutions: those studios where iterative practice can be facilitated (record-review-revise cycles), where creative tools are readily available, and where the setup and feedback processes are not hindered can reinforce performance self-efficacy, positively associated with creative musicianship. In practical terms, it means the focus on the reliable recording and playback options, easy access to the software, reliable connectivity, and the feature of always-available assistance that would enable students to make practice visible in terms of improvement. Since teacher autonomy support reinforced the efficacy mechanism, departments ought to combine studio upgrades with instructor development centered on autonomy-supportive behaviors, which are providing meaningful choice, taking into account student views, relying on informational (non-controlling) feedback and promoting self-initiated experimentation. This congruency is important: even powerful studio affordances may be underused when the students feel the climate to be controlling or risk-averse. Programs may also track self-efficacy in performance as a useful early measurement in assessment and advising of whether studio resources and teaching practices are creating the desired psychological conditions to facilitate creativity. Lastly, since mastery goal orientation did not reinforce the indirect path, practitioners should not believe that encouraging students to be mastery-oriented will be the only way to open the door to the advantages of studio technology. Rather, the more dependable leverage points seem to be (a) enhancing the quality of studio design and (b) developing autonomy-supportive instruction that might assist them internalize practice objectives and persevere through creative trial-and-error.

In the case of music departments and studios, the findings indicate that technology-enhanced studio design can be more advantageous in case of autonomy-supportive teaching practices. This might be achieved in practice through (a) designing studio spaces so that feedback and iteration are simple (recording/playback, visualization, modular setup), (b) designing assignments that promote experimentation and revision, and (c) training instructors to offer autonomy supports (choices, rationalizations, non-controlling feedback) so that students can use studio affordances as competence-building experiences. These implications are to be understood as evidence-based suggestions and not prescriptions because of the cross-sectional design.

5.3. Limitations and future research directions

Certain restrictions are to be highlighted. To begin with, the cross-sectional survey design is unable to determine a causal direction and exclude the possibility of mutual relationships. Second, self-report measures may be subject to common method variance bias; procedural remedial measures and checks of markers minimize the threat, but do not eliminate it. Third, the research was carried out within a certain cultural/educational situation, which might determine the practices of technology integration and the motivational situation, too. The next round of work ought to (a) adopt longitudinal or intervention designs,

(b) adopt behavioral or performance-based measures of creativity (to supplement self-reports), and (c) examine other boundary conditions, including performance anxiety, specificity of teacher feedback, and prior technology experience.

6. Conclusion

This study contributes to the study of music education because it explains the relationship between technology-enhanced studio design and creative musicianship and identifies the conditions, psychological and contextual, that influence this relationship. The results suggest that technology-enhanced studio design is positively related to creative musicianship and that this relationship works, in part, through performance self-efficacy. In other words, when the studio environments are supportive of iterative practice and experimentation with accessible tools and feedback-rich conditions, students are more likely to feel that they can meet performance demands, and students with this confidence are found to have stronger creative musicianship. The findings of the conditional process further indicate that teacher autonomy support strengthens the indirect pathway through performance self-efficacy, suggesting that it is important to align studio design with an autonomy supportive instructional climate. In contrast, mastery goal orientation did not significantly enhance the mediated pathway, which suggests that the influence of contextual supports may be more important than the influence of learner goal orientation in facilitating the translation of studio affordances into efficacy-building experiences and creative development. In sum, this study provides evidence that technology-enhanced studio design is positively associated with creative musicianship and that performance self-efficacy is a key explanatory mechanism. The findings also suggest that teacher autonomy support strengthens this mechanism, highlighting that motivational climate shapes how learners benefit from technology-rich learning environments. While the results are theoretically consistent with need-supportive approaches and creativity-supporting learning environments, future research using longitudinal or experimental methods is needed to confirm causal processes and to refine boundary-condition explanations.

CRediT authorship contribution statement

Shiyi Xiao: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Atif Saleem:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.actpsy.2026.107215>.

Data availability

Data will be made available on request.

References

- Ahmad, N., Ahmad, A., Lewandowska, A., & Han, H. (2024). From screen to service: How corporate social responsibility messages on social media shape hotel consumer advocacy. *Journal of Hospitality Marketing & Management*, 33(3), 384–413.
- Ahmad, N., Ahmad, A., & Siddique, I. (2024). Beyond self-interest: How altruistic values and human emotions drive brand advocacy in hospitality consumers through

- corporate social responsibility. *Corporate Social Responsibility and Environmental Management*, 31(3), 2439–2453.
- Ahmad, N., Samad, S., Ahmad, A., Han, H., & Raposo, A. (2026). Breaking new ground in hotel employee health and tourism in pursuit of sustainable development goals: A case study. *International Journal of Hospitality Management*, 133, Article 104493.
- Ahmad, N., Samad, S., & Han, H. (2024). Charting new terrains: How CSR initiatives shape employee creativity and contribute to UN-SDGs in a knowledge-driven world. *Journal of Innovation & Knowledge*, 9(4), Article 100557.
- Ahmad, N., Samad, S., & Mahmood, S. (2024). Sustainable pathways: The intersection of CSR, hospitality and the United Nations' sustainable development goals. *Current Issues in Tourism*, 1–20.
- Bao, H., & Yu, L. (2025). Performance anxiety, wellbeing, and cultural factors in Chinese music conservatories: A two-study research of student experiences. *Frontiers in Psychology*, 16, Article 1671941.
- Beattie, S., Woodman, T., Fakehy, M., & Dempsey, C. (2016). The role of performance feedback on the self-efficacy–performance relationship. *Sport Exercise and Performance Psychology*, 5(1), 1.
- Beik, A., & Cho, Y. (2024). Effects of goal orientation on online learning: A meta-analysis of differences in Korea and US. *Current Psychology*, 43(2), 1496–1506.
- Chen, M., Miao, Z., & Ahmad, N. (2026). Managing behavior in AI-mediated English education: The interplay of teachers, technology, and communication willingness. *Acta Psychologica*, 265, Article 106765.
- China Education Daily. (2024). Overview of work on digital education in China. In *China Education Daily.com*. Retrieved September 11, 2025 from http://en.moe.gov.cn/features/2024WorldDigitalEducationConference/News/202402/t20240201_1113777.html.
- Davies, D., Jindal-Snape, D., Collier, C., Digby, R., Hay, P., & Howe, A. (2013). Creative learning environments in education—A systematic literature review. *Thinking Skills and Creativity*, 8, 80–91.
- De Brabander, C. J., & Martens, R. L. (2014). Towards a unified theory of task-specific motivation. *Educational Research Review*, 11, 27–44.
- Edwards, K. D., & Konold, T. R. (2020). Moderated mediation analysis: A review and application to school climate research. *Practical Assessment, Research & Evaluation*, 25, 5.
- Ge, X. (2024). *Music teachers' perceptions of creativity in the context of twenty-first century Chinese education: A qualitative study on the influence of experience and policy on primary school music and piano teachers*. University of Glasgow.
- Haase, J., Hoff, E. V., Hanel, P. H., & Innes-Ker, Å. (2018). A meta-analysis of the relation between creative self-efficacy and different creativity measurements. *Creativity Research Journal*, 30(1), 1–16.
- Hair, J., & Alamer, A. (2022). Partial Least Squares Structural Equation Modeling (PLS-SEM) in second language and education research: Guidelines using an applied example. *Research Methods in Applied Linguistics*, 1(3), Article 100027.
- Hamond, L., Himonides, E., & Welch, G. (2020). The nature of feedback in higher education studio-based piano learning and teaching with the use of digital technology. *Journal of Music Technology & Education*, 13(1), 33–56.
- Hayes, A. F. (2017). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. Guilford publications.
- Hayes, A. F., & Rockwood, N. J. (2020). Conditional process analysis: Concepts, computation, and advances in the modeling of the contingencies of mechanisms. *American Behavioral Scientist*, 64(1), 19–54.
- Hendriks, L. H., Steenbeek, H. W., Bisschop Boele, E. H., & van Geert, P. L. (2023). *Promoting creative autonomy support in school music education: An intervention study targeting interaction*. Frontiers in Education.
- Huang, C. (2016). Achievement goals and self-efficacy: A meta-analysis. *Educational Research Review*, 19, 119–137.
- Imran, M., Xu, H., Wei, N., Alzuman, A., & Ahmad, N. (2026). Emotional and cognitive mechanisms of eco-social receptivity: Exploring tourism brand e-vocacy in AI-enabled travel. *Acta Psychologica*, 263, Article 106311.
- Jiang, M., Ahmad, N., Arshad, M. Z., & Liu, C. (2025). From digital corporate social responsibility (CSR) to consumer voice: shaping CSR-specific electronic word of mouth in hospitality industry. *Journal of Hospitality Marketing & Management*, 1–34.
- Kaufman, J. C. (2012). Counting the muses: development of the Kaufman domains of creativity scale (K-DOCS). *Psychology of Aesthetics, Creativity, and the Arts*, 6(4), 298.
- Kock, N. (2015). Common method bias in PLS-SEM: A full collinearity assessment approach. *International Journal of e-Collaboration (ijec)*, 11(4), 1–10.
- Lam, C. K. (2024). Technology-enhanced creativity in K-12 music education: A scoping review. *International Journal of Music Education*, 42(4), 691–703.
- Li, S. (2023). The effect of teacher self-efficacy, teacher resilience, and emotion regulation on teacher burnout: A mediation model. *Frontiers in Psychology*, 14, Article 1185079.
- Li, Y., Ahmad, N., & Lewandowska, A. (2024). Eco-Chic stays: The influence of green communication on guest booking intentions. *Journal of Vacation Marketing. Ahead of print*, 13567667241293792.
- Lindell, M. K., & Whitney, D. J. (2001). Accounting for common method variance in cross-sectional research designs. *Journal of Applied Psychology*, 86(1), 114.
- Liu, C., Ahmad, N., Arshad, M. Z., & Mingdi, J. (2026). Connecting green dots: the role of social media in sustainable tourism decision-making. *Journal of Hospitality and Tourism Technology*, 1–21. <https://doi.org/10.1108/jht-11-2024-0734>. Ahead of Print.
- Liu, M. (2025). Why we play: A motivational–efficacy pathway from basic psychological needs to creative musicianship. *Acta Psychologica*, 261, Article 105952.
- Macrides, E., & Angeli, C. (2020). Music cognition and affect in the design of technology-enhanced music lessons. *Frontiers in Education*, 5, Article 518209.

- Midgley, C., Maehr, M. L., Hruda, L. Z., Anderman, E., Anderman, L., Freeman, K. E., & Urdan, T. (2000). *Manual for the patterns of adaptive learning scales* (pp. 734–763). Ann Arbor: University of Michigan.
- Miller, D. S. (2022). Creativity in concert: Centering creative musicianship in the orchestra classroom. *American String Teacher*, 72(3), 17–21.
- Namazi, M., & Namazi, N.-R. (2016). Conceptual analysis of moderator and mediator variables in business research. *Procedia Economics and Finance*, 36, 540–554.
- Papaioannou, G., Volakaki, M.-G., Kokolakis, S., & Vouyioukas, D. (2023). Learning spaces in higher education: A state-of-the-art review. *Trends in Higher Education*, 2(3), 526–545.
- Ryan, C., Andrews, N., & Tsang, J. (2025). Music performance anxiety and perfectionism: A comparison of in-person and virtual contexts. *Musicae Scientiae*, 29(2), 330–345.
- Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68.
- Ryan, R. M., & Deci, E. L. (2020). Intrinsic and extrinsic motivation from a self-determination theory perspective: Definitions, theory, practices, and future directions. *Contemporary Educational Psychology*, 61, Article 101860.
- Sciences, C. N. A.o. E. (2023). *Report on China smart education 2022: Digital transformation of chinese education towards smart education*. Springer Nature.
- Tan, L., & Sin, H. X. (2020). Achievement goals in music contexts: A review of literature. *Update: Applications of Research in Music Education*, 38(2), 10–19.
- Tang, M., Wang, D., & Guerrien, A. (2020). A systematic review and meta-analysis on basic psychological need satisfaction, motivation, and well-being in later life: Contributions of self-determination theory. *PsyCh Journal*, 9(1), 5–33.
- Thaçi, F. N. (2025). The rise of technology in music education: A bibliometric study of a rapidly growing field. *Architecture Image Studies*, 6(4), 247–267.
- Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 425–478.
- Waddell, G., & Williamon, A. (2019). Technology use and attitudes in music learning. *Frontiers in ICT*, 6, 11.
- Waddington, J. (2023). Self-efficacy. *ELT Journal*, 77(2), 237–240.
- Wang, Q., Gao, Y., & Sun, J. (2025). What matters most? AQ-Method study on Chinese university students' perceived teacher support in AI-assisted EFL learning. *European Journal of Education*, 60(4), Article e70327.
- Wang, Q., Gao, Y., & Wang, X. (2025). Exploring engagement, self-efficacy, and anxiety in large language model EFL learning: A latent profile analysis of Chinese university students. *International Journal of Human-Computer Interaction*, 41(12), 7815–7824.
- Wang, R., Ahmad, N., Ryu, H. B., Comite, U., Ariza-Montes, A., & Han, H. (2024). Healing leaders: Altruism and psychological safety as antidotes to burnout in health care settings. *Social Behavior and Personality: An International Journal*, 52(12), 1–11.
- Wang, Y., & Guo, Y. (2026). Investigating the interplay of Chinese english as foreign language learners' growth mindset, enjoyment, adaptability, and engagement in AI-mediated informal digital learning of english. *Journal of Computer Assisted Learning*, 42(1), Article e70189.
- Williams, G. C., Lynch, M. F., McGregor, H. A., Ryan, R. M., Sharp, D., & Deci, E. L. (2006). Validation of the "important other" climate questionnaire: Assessing autonomy support for health-related change. *Families, Systems & Health*, 24(2), 179.
- Xu, S., & Li, D. (2025). Autonomy-supportive music teaching, collective learning, flow, and music Students' well-being: A mediational model. *Acta Psychologica*, 254, Article 104827.
- Yang, Y., & Welch, G. (2023). A systematic literature review of Chinese music education studies during 2007 to 2019. *International Journal of Music Education*, 41(2), 175–198.
- Yim, E. P.-Y. (2023). Self-efficacy for learning beliefs in collaborative contexts: Relations to pre-service early childhood teachers' vicarious teaching self-efficacy. *Frontiers in Education*, 8, Article 1210664.
- Zelenak, M. S. (2015). Measuring the sources of self-efficacy among secondary school music students. *Journal of Research in Music Education*, 62(4), 389–404.
- Zimu, Y. (2024). Examining the relationship between teacher self-efficacy and student engagement in technology-enhanced learning environments. *Development*, 6(2), 115–119.